Santiago Montaña

Data Scientist | Data & AI Engineer | Conversational AI & Cloud Operations

Bogotá / Boyacá, Colombia · Hybrid / Remote • santiloc23456@gmail.com • +57 310 550 8171

• https://www.linkedin.com/in/santiago-montaña/ • https://santiloc-hub.github.io/santiloc-hub/

PROFESSIONAL SUMMARY

Data Scientist and Data & AI Engineer with a background in Electronic Engineering and a Specialization in Strategic Data Analytics (UPTC). Experienced in end-to-end data pipeline engineering (Snowflake, Apache Airflow, dbt, Meltano), predictive modeling and classification algorithms (Random Forest, XGBoost, Monte Carlo quantitative simulations, MLflow), conversational AI solutions (LLMs, autonomous agents, Model Context Protocol - MCP, ElevenLabs voice AI), and cloud infrastructure provisioning on AWS using Terraform (IaC), Docker, and FastAPI under CI/CD.

PROFESSIONAL EXPERIENCE

Junior Data Scientist & AI Developer

Jan 2026 – Present

CIM DATA IA — Consultoría de Datos e Inteligencia Artificial — Boyacá / Remoto, Colombia

- Developed predictive models and classification algorithms in Python (scikit-learn, Random Forest, XGBoost) for operational forecasting and prioritization, tracking experiments with MLflow.
- Built conversational AI solutions and autonomous agents integrating Model Context Protocol (MCP) to automate enterprise workflows and infrastructure deployment.
- Collaborated on cloud infrastructure provisioning and deployment on AWS (EC2, S3, IAM) using Terraform (IaC), ensuring reproducible and scalable environments.
- Executed quantitative Monte Carlo simulations and stress tests to validate data design effectiveness, orchestrating ETL/ELT pipelines with Python, SQL, Snowflake, dbt, and Apache Airflow under CI/CD.

Development Assistant (Data & AI)

May 2025 – Dec 2025

CIM DATA IA — Boyacá, Colombia

- Analytically optimized and tuned banking chatbots and conversational AI flows, measurably improving intent classification accuracy and client response quality.
- Collaborated on building an AI voice assistant for automated loan verification in the financial domain using Python, ElevenLabs, and third-party REST APIs.
- Implemented operational process automations and standardized experimentation environments using Anaconda and virtualenv to guarantee cross-team portability.

Data & Industrial Operations Automation Intern

May 2024 – Nov 2024

HOLCIM S.A. — Operaciones Industriales y Planta — Nobsa, Boyacá, Colombia

- Automated collection, cleaning, and processing of industrial plant operational data using Python scripts and Google Apps Script.
- Designed and implemented a centralized database to improve industrial asset traceability, reliability, and organization.
- Built interactive executive dashboards in Looker Studio to monitor plant productivity, equipment availability, and operational KPIs.

TECHNICAL SKILLS

Data Engineering & Orchestration:	Snowflake (Cloud Data Warehouse), Apache Airflow (DAG Orchestration), dbt (SQL Transformations), Meltano (ELT Extraction), Pipelines ETL / ELT, Modelamiento de Datos & Calidad
Machine Learning & Data Science:	Scikit-learn (Random Forest, XGBoost, Ensembles), Simulaciones Cuantitativas (Monte Carlo), MLflow (Experiment Tracking & MLOps), TensorFlow & PyTorch, Análisis Estadístico & Validación de Hipótesis, Modelado Predictivo & Clasificación
Conversational AI & MCP Agents:	Model Context Protocol (MCP) Integration, Agentes Autónomos de IA, LLMs (Claude, OpenAI, Prompt Engineering), ElevenLabs (Asistentes de Voz con IA), Optimización de Chatbots Bancarios & NLP
Cloud, DevOps & IaC:	Terraform (Infrastructure as Code - IaC), AWS (EC2, S3, IAM), FastAPI & Pydantic (APIs REST en Producción), Docker & Entornos Reproducibles (Anaconda, virtualenv), CI/CD con GitHub Actions & Git
Programming Languages:	Python (pandas, NumPy, SciPy, scikit-learn, Matplotlib), SQL (Consultas Analíticas, CTEs, Optimización), R (Estadística & Visualización), JavaScript / TypeScript (ES6+), C++
Databases & Warehousing:	Snowflake, PostgreSQL, MySQL, SQLite

Visualization & BI: Looker Studio, Power BI, Tableau, Storytelling con Datos & KPIs Operativos

FEATURED PROJECTS

Autonomous News & Media AI Agent *Stack: Python, LLMs, MCP, Autonomous Agents, Automation*

Autonomous AI pipeline integrating LLMs and structured workflows for automated news research, synthesis, and multi-channel content generation.

Multiclass Vehicle Classification API & Web App *Stack: Python, scikit-learn, FastAPI, Flask, Streamlit, Pydantic*

End-to-end vehicle classification model serialized and deployed with FastAPI/Flask and Streamlit, including Pydantic validation, logging, and automated testing.

USD/COP Exchange Rate (TRM) & Crypto Monte Carlo Predictor *Stack: Python, Scikit-learn, Monte Carlo, NLP Sentiment, Jupyter Notebook*

Predictive modeling combining historical time series, real-time news sentiment analysis, and Monte Carlo quantitative simulations to estimate market scenarios.

Conversational Voice AI Agent (ElevenLabs + LLMs) *Stack: Python, ElevenLabs, LLMs, Voice AI, REST APIs*

Interactive voice AI agent built with Python and ElevenLabs APIs for low-latency automated verification and natural conversational workflows.

EDUCATION

Specialization in Strategic Data Analytics
Universidad Pedagógica y Tecnológica de Colombia (UPTC)

Bachelor of Science in Electronic Engineering
Universidad Pedagógica y Tecnológica de Colombia (UPTC)

LANGUAGES

Spanish — Native (C2)

English — Upper-Intermediate (B2 professional reading & technical documentation)

CERTIFICATIONS & COURSES

MLOps & Despliegue de Modelos de Machine Learning	2025	Data Scientist con Python	Apr 2025
Decision Trees y Random Forest con scikit-learn	Mar 2025	Tableau — Visualización de Datos y Storytelling	Mar 2025
Regresión Lineal con Python y scikit-learn	Feb 2025	Crear tus Proyectos de Ciencia de Datos	Feb 2025
Introducción a Machine Learning	Jan 2025	Cálculo Diferencial para Data Science	Jan 2025
Git y GitHub Profesional	Jan 2025	Estadística Descriptiva para Data Science	Dec 2024
Jupyter Notebooks y Anaconda para Ciencia de Datos	Nov 2024	Bases de Datos con SQL	Sep 2024
Python para Ciencia de Datos	Sep 2024	Power BI para Analítica de Negocios	2025